

BUSINESS SECURITY ALLIANCE

Full Platform Documentation

User Manual · Developer Guide · System Architecture · Administration Guide

Next.js 14 · TypeScript · PostgreSQL · Prisma · Tailwind CSS

Document Version 1.0

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1. Executive Overview & Platform Purpose

The Business Security Alliance (BSA) is a professional association platform designed specifically for the global security industry. Unlike generic corporate brochure websites, BSA is a dynamic, membership-driven network that connects security leaders, cybersecurity directors, physical defense specialists, enterprise risk officers, and security vendors.

The Five Core Pillars

- **CONNECT** — Help security practitioners and executives locate peers via a searchable, privacy-governed Member Directory.
- **GROW** — Facilitate business growth, career advancement, and strategic partnerships via the Opportunities Board.
- **LEARN** — Provide practitioner-written playbooks, CPD-credited educational modules, and recorded executive roundtables.
- **PARTICIPATE** — Engage members through live events, workshops, regional chapter meetups, and community discussions.
- **BUILD THE INDUSTRY** — Advance converged physical and cyber security resilience across enterprises worldwide.

2. Technology Stack & Architecture

BSA is built on a modern, high-performance, serverless-ready stack:

Layer	Technology	Key Details
Framework	Next.js 14 (App Router)	Server-side rendering (SSR), Static Generation (SSG), Incremental Static Regeneration (ISR), and React Server Components (RSC).
Language	TypeScript 5.x	End-to-end static typing across database models, API payloads, and client UI props.
Styling & System	Tailwind CSS + Vanilla CSS Tokens	Custom design system using dark mode, glassmorphism, steel/silver metallic accents, and responsive layout primitives.
Database	PostgreSQL (Supabase)	Managed PostgreSQL with dual connection pool configuration.
ORM	Prisma ORM 5.x	Type-safe query builder with automated schema migrations.
Connection Pooler	PgBouncer (Transaction Mode)	Port 6543 used for runtime lambdas (DATABASE_URL) with connection_limit=10 and pool_timeout=20. Port 5432 used for Prisma CLI migrations (DIRECT_URL).
Authentication	JWT + Bcrypt	Cryptographically signed bsa_session cookies stored as HttpOnly, SameSite=Lax/Strict, and Secure. Passwords salted with bcryptjs.
Validation	Zod 3.x	Input sanitization, email verification, and payload coercion.
Icons & Media	Phosphor Icons (SSR)	Server-side rendered icon set (@phosphor-icons/react/dist/ssr).

3. User Roles & Access Levels

The platform enforces three distinct user roles:

1 • GUEST (Unauthenticated Visitor)

- Can browse public marketing pages, blog insights, event listings, chapter radar, resource overviews, and sponsorship info.
- Can register for public events, submit contact inquiries, and apply for membership.
- Cannot view private member directory details, access member dashboard, join chapters, or participate in peer chat.

2 • MEMBER (Authenticated Security Professional)

- Has full access to the Member Dashboard (/dashboard).
- Can customize public/private profile fields, specialties, and contact preferences.
- Can view the full Member Directory, search members by discipline, and initiate 1-on-1 direct messaging.
- Can join regional chapters, claim event tickets, track CPD learning credits, and apply for job/board opportunities.

3 • ADMIN (System Administrator)

- Has exclusive access to the Admin Console (/admin).
- Governed by the Single Primary Admin Policy (admin@bsa.in).
- Full write/delete capabilities across all entities (members, events, chapters, opportunities, posts, resources, broadcasts, CRM leads, and custom scripts).

4. Core Feature Modules & User Experience

4.1 Member Directory & Public Profiles · [/directory](#), [/members/\[handle\]](#)

- **Search & Filters:** Search members by name, organization, job title, handle, or location. Filter by role, discipline, or member type (PROFESSIONAL, LEADER, CONSULTANT, VENDOR, ORGANISATION).
- **Granular Privacy Matrix:** Each member independently controls visibility settings — isPublic, searchableInDirectory, and per-field toggles for showEmail, showPhone, showOrg, showLinkedIn, and showWebsite.

4.2 Events & Ticketing Programme · [/events](#), [/events/\[slug\]](#)

- **Event Categories:** WEBINAR, WORKSHOP, ROUNDTABLE, SUMMIT, NETWORKING.
- **Ticketing & Seat Capacity:** Visual real-time capacity meters (green <60%, orange 60–90%, red >90%). Supports free or paid ticket registrations.
- **CPD Hours:** Earn Continuing Professional Development (CPD) credits upon attending certified events.
- **Registration Code:** Generates a unique registration code for every attendee.

4.3 Regional Chapters & Interactive 3D Radar · [/chapters](#), [/chapters/\[slug\]](#)

- **Interactive 3D Globe Radar:** A WebGL/Canvas 3D globe plotting active regional chapters using real latitude and longitude coordinates.
- **Chapter Roster:** View chapter leads, chairs, meeting cadences (Monthly, Bi-Monthly, Quarterly), and local members.
- **Join Flow:** Members can join regional chapters with one click.

4.4 Business Opportunities & Applications · [/opportunities](#), [/opportunities/\[slug\]](#)

- **Listings Board:** Discover executive roles, consulting tenders, speaking slots, board seats, and corporate partnerships.
- **Direct Application:** Submit applications with attached resume links, cover notes, and experience summaries.

4.5 Knowledge Playbooks & Learning Modules · [/resources](#), [/resources/\[slug\]](#)

- **Executive Frameworks:** Structured playbooks covering converged physical/cyber risk management, AI surveillance compliance, and incident response.
- **Module Progress Tracker:** Track completed modules and earn completion certificates.

4.6 Published Thinking & Insights (Blog) · [/blog](#), [/blog/\[slug\]](#)

- **Industry Articles:** Searchable repository of articles written by BSA leaders and security executives.
- **Featured Article:** Prominent hero placement for primary featured insights.

4.7 Industry Partners & Sponsorships · [/sponsors](#)

- **Sponsorship Tiers:** DIAMOND, GOLD, SILVER, COMMUNITY.
- **Partner Badges:** Custom perks, hiring indicators (Is Hiring), and direct website CTA links.

4.8 Real-Time Peer Messaging & Community Chat · /community

- **Channels & DMs:** Public channel chat (#general) and private 1-on-1 direct messaging between members.
- **Read Receipts (ChatRead):** Persistent high-water mark tracking showing when peers have read messages.
- **Rich Media:** Supports image attachments and emoji reactions.

4.9 Site Broadcasts & In-App Notifications ·

- **Targeted Notification Banners:** Display site-wide announcements, security advisories, and platform updates.
- **Targeting:** Route alerts to ALL users, MEMBER accounts, or specific individuals.

5. Admin Console Command Center (/admin)

The Admin Console is the central control system for managing the entire platform.

5.1 Single-Admin Governance Policy

To guarantee operational stability, the platform enforces a strict single-admin rule:

- Only **admin@bsa.in** holds administrative rights.
- Creation of additional admin accounts or demotion of admin@bsa.in is programmatically blocked in backend API routes.

5.2 Live Metrics Dashboard

Displays real-time KPIs:

- Total Members & 30-Day Growth Rate
- Total Active Events & Registrations
- Open CRM Leads / Form Inquiries
- Chapter Counts & Active Resources

5.3 Member Management & Access Control

- Review pending activation requests.
- Activate or revoke member access.
- Filter members by status (ACTIVE, PENDING, REVOKED) or role.

5.4 Event Management

- Create, edit, or delete events.
- Update prices or ticket details safely. When editing an event, the system uses a **selective patch diff engine** (changedFields) to update only changed fields, leaving untouched attributes intact.

5.5 Regional Chapter Radar Management

- Add new chapters with city, region, coordinates (latitude/longitude), meeting cadence, and accent theme.
- Toggle chapter status (Active / Deactivate).

5.6 Opportunities & Application Tracking

- Create new role/tender listings.
- Review submitted candidate applications.

5.7 Broadcast Notifications Panel

- Create, pin, or deactivate site-wide notifications (ANNOUNCEMENT, ADVISORY, UPDATE).

5.8 Enquiries & Leads Inbox (HubSpot CRM Sync)

- Review form submissions (Contact, Sponsorship, Chapter Inquiries).
- Tracks synchronization status with HubSpot CRM (PENDING, SYNCED, FAILED).
- One-click reply links (mailto:) and status toggle (Mark Handled).

5.9 Custom Scripts & Code Injector CMS (ScriptsPanel)

Inject custom third-party scripts dynamically from the UI without redeploying code:

- **Header Scripts (<head>):** Google Analytics (GA4), Meta Pixel, Google Tag Manager.
- **Footer Scripts (</body>):** Live chat widgets (Crisp, Intercom, Drift).
- **Page-Specific Body Injections:** Target scripts to specific pathnames (e.g. /contact, /pricing).
- **Templates:** One-click code insertion templates for popular integrations.

5.10 Media Asset Directory Manager

- Upload and manage global asset URLs, images, and brand files.

6. Security, Protection & Compliance Architecture

The BSA platform is engineered to defend against common web security threats:

BSA SECURITY ARCHITECTURE	
[Layer 1: Transport]	- HTTPS / Enforced TLS (sslmode=require)
[Layer 2: Network]	- IP Rate Limiting (Token Bucket)
[Layer 3: Authentication]	- HttpOnly Signed JWT Cookie + Bcrypt
[Layer 4: Authorization]	- Server-Side RBAC Guards (requireAdmin)
[Layer 5: Input Guard]	- Strict Zod Schema Validation & Trimming
[Layer 6: Data Layer]	- Prisma ORM Parameterized Queries (SQLi)
[Layer 7: Output Guard]	- React Automatic JSX Escaping (XSS)

1. SQL Injection (SQLi) Immunity

All database operations use Prisma ORM parameterized queries. Inputs are treated strictly as literal values by PostgreSQL.

2. Cross-Site Scripting (XSS) Protection

React automatically escapes strings rendered in JSX templates.

3. Session Hijacking Defense

Session tokens (bsa_session) are stored in HttpOnly, SameSite, Secure cookies, preventing client-side script access.

4. Brute-Force Protection

Rate limiting middleware (src/lib/rate-limit.ts) restricts authentication endpoints — 5 attempts/min on login, 3 attempts/min on register.

5. No-Index Admin Privacy

All /admin routes include <meta name="robots" content="noindex, nofollow"> to exclude administrative tools from search engine indexes.

7. Database Schema & Data Model

Defined in prisma/schema.prisma:

Entity	Description
User	Account identity, email, password hash, role (ADMIN, MEMBER), status (ACTIVE, PENDING, REVOKED).
MemberProfile	Full name, handle, job title, headline, org, field, member type, bio, experience, skills, specialties.
MemberPrivacy	Granular visibility settings (showEmail, showPhone, showOrg, showLinkedIn, searchableInDirectory).
Chapter	Name, slug, region, city, country, latitude, longitude, cadence, active flag.
ChapterMembership	Connects users to chapters with roles (CHAIR, MEMBER).
Event	Title, slug, category, status, dates, times, location, capacity, CPD hours, price.
EventTicket	Ticket tiers, price, available/sold quantities.
EventRegistration	Connects users/attendees to events with unique registration codes.
Opportunity	Career/tender listings, org, requirements, deadline.
Application	Candidate submissions attached to opportunities.
BlogPost	Title, slug, content, category, tags, read time, featured status.
Resource	Knowledge playbooks and modules.
ResourceModule	Individual modules with sort order and content.
UserModuleProgress	Tracks module completion per member.
ChatMessage	Peer messaging content, channel, recipient, attachments.
ChatRead	Read receipts per user per conversation (ch:general or dm:<userId>).
FormSubmission	Contact and inquiry leads tracked for HubSpot CRM.
SiteNotification	Broadcast banners with role targeting.
CustomScript	CMS table storing header, footer, and page-body code injections.
MediaAsset	Directory of uploaded files and images.

8. API Route Sitemap

Endpoint	Method	Access	Description
/api/auth/login	POST	Public	Authenticates credentials and sets bsa_session cookie.
/api/auth/register	POST	Public	Registers a new member account.
/api/auth/logout	POST	Authenticated	Clears bsa_session cookie.
/api/auth/forgot-password	POST	Public	Generates password reset tokens.
/api/auth/reset-password	POST	Public	Consumes reset token and sets new password.
/api/user/profile	GET / PATCH	Member	Reads or updates current member's profile.
/api/user/privacy	PATCH	Member	Updates member directory visibility matrix.
/api/events/[id]/register	POST	Public / Member	Registers attendee for an event and returns registration code.
/api/chapters/[slug]/join	POST	Member	Joins a regional chapter.
/api/opportunities/[slug]/apply	POST	Public / Member	Submits an application for an opportunity.
/api/forms	POST	Public	Handles general contact, sponsorship, and chapter inquiries.
/api/chat/messages	GET / POST	Member	Fetches or sends channel/DM chat messages.
/api/chat/conversations	GET	Member	Lists active conversations and unread indicators.
/api/notifications	GET	Public / Member	Fetches active site notifications.
/api/scripts	GET	Public	Serves header/footer/body custom scripts for injection.
/api/admin/members	PATCH / DELETE	Admin	Activates, revokes, or deletes member accounts.
/api/admin/events	POST / PATCH / DELETE	Admin	Creates, edits, or deletes events and ticket prices.
/api/admin/chapters	POST / PATCH / DELETE	Admin	Manages regional chapters and 3D radar coordinates.
/api/admin/opportunities	POST / PATCH / DELETE	Admin	Manages career and business opportunity listings.
/api/admin/notifications	POST /	Admin	Manages broadcast notifications and

Endpoint	Method	Access	Description
	PATCH / DELETE		pinned advisories.
/api/admin/leads	PATCH / DELETE	Admin	Manages contact leads and HubSpot CRM status.
/api/admin/scripts	PATCH	Admin	Updates global custom scripts (Header/Footer/Body CMS).

9. Local Development & Deployment Guide

Prerequisites

- Node.js 18.x or 20.x
- npm 9.x or later
- Supabase PostgreSQL Database (or local PostgreSQL)

Local Setup Instructions

1. Clone & Install Dependencies

```
git clone https://github.com/bsawebsitedesign-afk/bsa.git
cd bsa
npm install
```

2. Configure Environment Variables

Copy .env.example to .env and fill in credentials:

```
DATABASE_URL="postgresql://postgres.ref:[PASSWORD]@aws-0-ap-south-1.pooler.supabase.com:6543/postgres?pgbouncer=true&connection_limit=10&pool_timeout=20"
DIRECT_URL="postgresql://postgres.ref:[PASSWORD]@aws-0-ap-south-1.pooler.supabase.com:5432/postgres"
JWT_SECRET="your-strong-random-jwt-secret-key"
NEXT_PUBLIC_APP_URL="http://localhost:3000"
```

3. Initialize Database & Seed Data

```
npx prisma db push
npm run db:seed
```

4. Run Local Development Server

```
npm run dev
```

Open <http://localhost:3000> in your browser.

5. Test Build Verification

```
npm run build
```

Default Credentials (Seed Data)

Account	Email	Password	Name
Primary Admin	admin@bsa.in	Admin@1011	—
Demo Member 1	member1@bsa.in	Member@1011	Marcus Vance
Demo Member 2	member2@bsa.in	Member@1011	Elena Rostova

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